



COLORECTAL CANCER

▪ The AIM of this lecture is

- 1- Define Colorectal cancer
- 2- know the etiology of Colorectal cancer
- 3- Identify the clinical features of Colorectal cancer
- 4- know how to diagnose Colorectal cancer
- 5- Discusses the treatment of Colorectal cancer

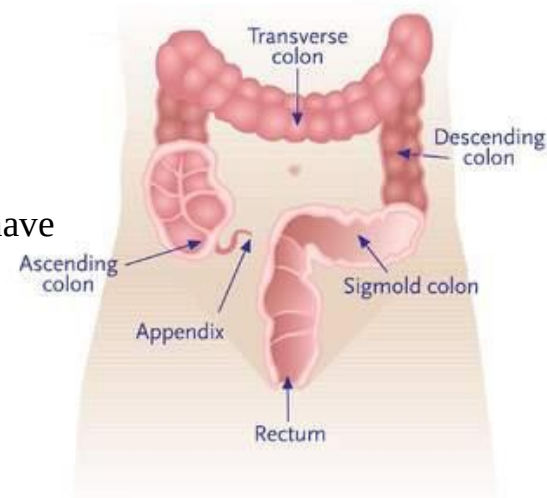
▪ Intended learning outcomes:

By the end of this lecture the student will be able to:

- 1- Recognize the clinical features of Colorectal cancer
- 2- Interpret the clinical data and investigation
- 3- Diagnose lymphoma
- 4- Arrange treatment plan for patients with Colorectal cancer

Colorectal cancer

- Globally, colorectal cancer is the third most commonly diagnosed cancer in males and second in females.
- High-income countries have the highest incidence of colorectal cancer, while Africa and South Asia have the lowest incidence.
- Worldwide incidence is continuing to increase.
- Men are affected slightly more commonly than women.
- In Iraq, it is the 5th commonest cancer.
- It occurs predominantly in the elderly, but the incidences have increased in the younger population .





Pathophysiology

- Both environmental and genetic factors are important in colorectal carcinogenesis.
- 75% of colorectal cancers are sporadic,
- 20% of patients have genetic predisposition,
- 5% being hereditary:
 - Non-polyposis:
 - Hereditary non-polyposis colorectal cancer HNPCC.
 - Polyposis:
 - Familial adenomatous polyposis FAP
 - Peutz–Jeghers syndrome
 - Juvenile polyposis

Familial adenomatous polyposis (FAP)

- Familial adenomatous polyposis (FAP) is the most.
- Common adenomatous polyposis syndrome.
- It is an autosomal dominant inherited disorder characterized by the early onset of hundreds to thousands of adenomatous polyps throughout the colon.
- If left untreated, all patients with this syndrome will develop colon cancer by age 35-40 years.
- In addition, an increased risk exists for the development of other malignancies.
- FAP has been described in all races, and males and females are equally affected (1:1).
- The average age of onset of polyposis in FAP is 16 years, whereas the average age of onset for colorectal cancer is 39 years.

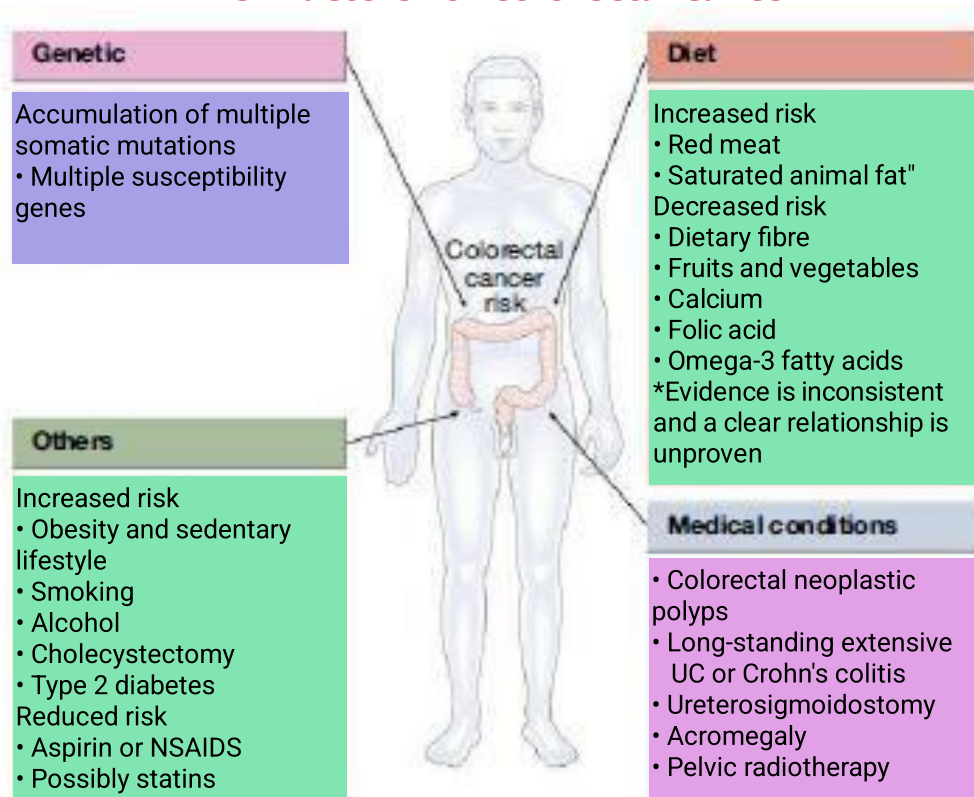




Site of colorectal tumors

- The most frequently affected sites:
- Rectum (38%)
- Sigmoid colon (20)
- Descending colon (10)
- Cecum (8%)
- Hepatic flexure (8%)

Risk factors for colorectal cancer



Genetics

- Colorectal cancer development results from the accumulation of multiple genetic mutations.
- There are also associated epigenetic influences, such as microRNA expression signature, and potential influences from non-coding genetic variation.
- There are three main pathways of genetic instability and each is associated with histological, clinical and prognostic parameters:
 - **Chromosomal instability**
 - **Microsatellite instability**
 - **CpG island methylator phenotype (CIMP).**



Pathways of carcinogenesis

- The classical adenoma–carcinoma pathway is characterised by an adenoma as the precursor lesion.
- Adenomas tend to result in carcinoma through the sequential acquisition of mutations, through chromosomal and microsatellite instability.
- The time period for an adenoma to develop into a carcinoma is thought to take between 5 and 10 years.
- The serrated neoplasia pathway is characterized by a serrated polyp as a precursor lesion.
- Serrated polyps (sessile serrated lesions, traditional serrated adenomas) were all thought to be non-neoplastic previously.
- However, it is now thought that sessile-serrated lesions and traditional serrated adenomas are precursor lesions for malignancy.

Clinical features

- Symptoms vary, depending on the site of the carcinoma. However, the disease is largely asymptomatic until at an advanced stage.
- In tumors of the left colon, fresh rectal bleeding is common and obstruction occurs early.
- Tumors of the right colon present with anemia from occult bleeding or with altered bowel habits, but the obstruction is a late feature.
- Colicky lower abdominal pain is present in two-thirds of patients and rectal bleeding occurs in 50%.
- A minority present with features of either obstruction or perforation, leading to peritonitis, localized abscess or fistula formation.
- Carcinoma of the rectum usually causes early bleeding, mucus discharge or a feeling of incomplete emptying.
- Between 10% and 20% of patients present with iron deficiency anaemia or weight loss.
- On examination, there may be a palpable mass, signs of anaemia or hepatomegaly from metastases.
- Low rectal tumours may be palpable on digital examination.



Investigations

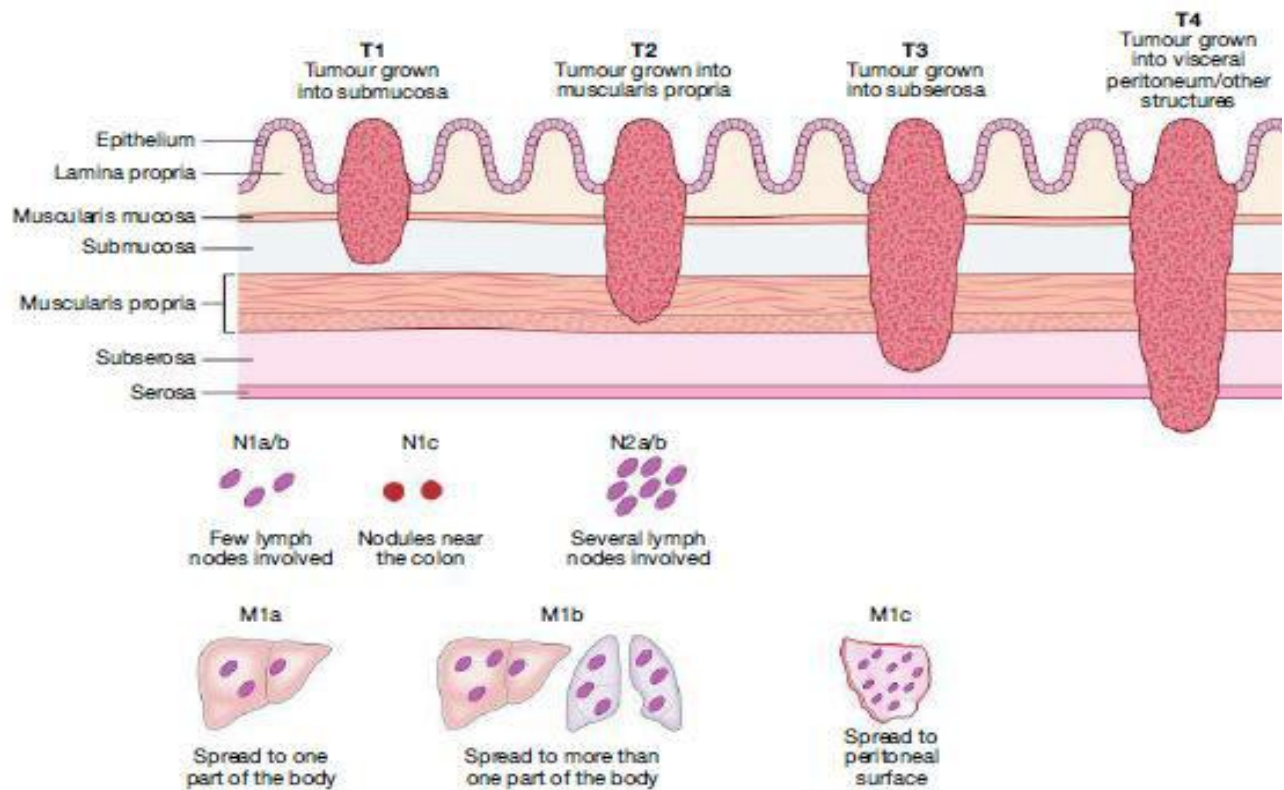
- **Colonoscopy** is the investigation of choice because it is more sensitive and specific than barium enema. Furthermore, lesions can be biopsied and polyps removed.
- **Virtual colonoscopy:** Patients in whom colonoscopy is incomplete and those who are at high risk of complications can be investigated by CT colonography.
- This is a sensitive and non-invasive technique for diagnosing tumours and polyps of more than 6mm diameter.
- **Imaging test:** When the diagnosis of colon cancer has been made, CT of the chest, abdomen, and pelvis should be performed as a staging investigation, particularly to detect hepatic metastases, with an increasing role of MRI for further characterization of liver lesions. Pelvic MRI or endoanal ultrasound should be used for local staging of rectal cancer, FDG PET-CT is being increasingly used.
- **Tumor marker:** Measurement of serum carcinoembryonic antigen (CEA) levels is of limited value in diagnosis, since values are normal in many patients, but CEA testing can be helpful during follow-up to monitor for recurrence.



Virtual colonoscopy



Staging



Staging

Table. Colon and Rectum Cancer Staging⁶

AJCC Stage	TNM Stage	Description
0	Tis N0 M0	Tumor is confined to mucosa
I	T1 N0 M0	Tumor invades submucosa
I	T2 N0 M0	Tumor invades muscularis propria
IIA	T3 N0 M0	Tumor invades subserosa or beyond, no other organs involved
IIB	T4 N0 M0	Tumor invades adjacent organs or perforates visceral peritoneum
IIIA	T1-2 N1 M0	Metastasis to 1-3 regional lymph nodes with tumor invasion of submucosa and/or muscularis
IIIB	T3-4 N1 M0	Metastasis to 1-3 regional lymph nodes with tumor invasion of subserosa or adjacent organs
IIIC	Any T, N2 M0	Metastasis to 4 or more lymph nodes
IV	Any T, any N, M1	Metastasis to distant organs

Abbreviations: AJCC, American Joint Committee on Cancer; Tis, tumor (carcinoma) in situ.



Assessment and treatment of colorectal polyps

- Colonic polyps are commonly found incidentally during endoscopy.
If a polyp is found during sigmoidoscopy, a colonoscopy should be subsequently performed, as up to 40%–50% of affected individuals have further polyps more proximally in the colon.
- As neoplastic colonic polyps have malignant potential, these should be removed via colonoscopic polypectomy, as this considerably reduces subsequent colorectal cancer risk.
- When cancer cells are found within 2 mm of the resection margin of the polyp, when the polyp cancer is poorly differentiate or when lymphatic invasion is present, segmental colonic resection is recommended because residual tumour or lymphatic spread (in up to 10%) may be present.
- Malignant polyps without these features can be followed up by surveillance colonoscopy.

Management

Endoscopy:

- Early colorectal cancers may be amenable to endoscopic treatment, using techniques such as endoscopic mucosal resection, endoscopic submucosal dissection, and endoscopic full-thickness resection.
- In addition to such a procedure, surgical mesenteric lymphadenectomy may be required.

Surgery

- Surgery for colorectal cancer remains the cornerstone of treatment with curative intent.
- Colorectal cancer may present as an emergency with obstruction or perforation
- Obstruction can be treated by either a decompressing colostomy or endoscopic stenting.
- Metastatic disease confined to the liver or lung should be considered for resection, as this can be potentially curative if there is truly no metastasis.
- Neoadjuvant radiotherapy or chemoradiotherapy should be offered to individuals with high-risk rectal cancers to increase the subsequent chance of a complete (R0) surgical resection.
- Metastatic disease confined to the peritoneum can be considered for cytoreductive surgery and hyperthermic intraperitoneal chemotherapy, which shows benefits in survival in a subset of patients.
- Postoperatively, patients should undergo colonoscopy after 1 year and then 3 years later to search for local recurrence or development of new lesions.



Adjuvant therapy

- About 30%–40% of patients have lymph node involvement at presentation and are therefore at risk of recurrence.
- Most recurrences occur within 3 years of diagnosis and affect the liver, lung, distant lymph nodes, and peritoneum.
- Adjuvant chemotherapy with 5-fluorouracil/folinic acid or capecitabine, preferably in combination with oxaliplatin, can reduce the risk of recurrence

Rectal tumor

- Rectal cancers can be complex to operate on due to the anatomy of the pelvis.
- Neoadjuvant concurrent chemoradiotherapy is the standard of care for stage II and stage III rectal cancer.
- Following neoadjuvant therapy, the response to treatment should be assessed, using digital rectal examination (when the site is within reach), endoscopy, and a pelvic MRI performed using a rectal cancer protocol; endoscopic biopsy in this setting has a high false negative rate, which limits its usefulness in assessing for residual disease
- In selected patients with a clinical complete response, experienced centers with established protocols can offer a watch-and-wait strategy, which should include both routine surveillance for disease recurrence and assessment for local tumor regrowth

Palliation of advanced disease

- Surgical resection of the primary tumor is appropriate for some patients with metastases to treat obstruction, bleeding, or pain.
- RAS and BRAF V600E mutations are tested for in metastatic colorectal cancer in individuals being considered for systemic anti-cancer treatment, with outcomes worse if these mutations are present.

Palliative chemotherapy with 5-fluorouracil/folinic acid, capecitabine, oxaliplatin, or irinotecan improves survival.

- Patients with advanced metastatic disease may be treated with monoclonal antibodies such as panitumumab or cetuximab, which bind to and inhibit the epidermal growth factor receptor.



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- They are often given together with chemotherapy.
- Pelvic radiotherapy is sometimes useful for distressing rectal symptoms, such as pain, bleeding or severe tenesmus.
- Endoscopic laser therapy or insertion of an expandable metal stent can be used to relieve obstruction.

Prevention and screening

- Secondary prevention aims to detect and remove lesions at an early or pre malignant stage by screening the asymptomatic general population.
- Several potential methods exist:
- Population-based screening of people between the age of 50 to 74 years by regular fecal immunochemical testing (FIT) reduces colorectal cancer mortality and increases the proportion of early cancers detected.
- **Colonoscopy** remains the gold standard and allows preventative polypectomy, but it is expensive, requires bowel preparation, and carries risks (perforation approximately 1:1000). Many countries lack the resources to offer this form of screening.
- **Flexible sigmoidoscopy** is an alternative option and has been shown to reduce overall colorectal cancer mortality by approximately 35% (70% for cases arising in the rectosigmoid). It is recommended as a one-off test in individuals aged 55 years, with subsequent FIT from age 60 years.
- **CT colonography** is fast and low-risk and offers equivalent sensitivity to colonoscopy. Disadvantages include reduced sensitivity to detect polyps of less than 6 mm, the requirement for bowel preparation, exposure to ionizing radiation and its inability to offer therapeutic intent

References and recommended further readings:

- Davidsons Principles and Practice of Medicine.
- Medscape website